

Car Price Prediction

Milestone 4 — Further Improvements

INTRODUCTION AND OVERVIEW

Two directions of advancement are presented, built on the Milestone 3 results. **RANDOM_SEED = 42** throughout for full reproducibility. The M3 benchmark is **Ridge Regression: RMSE = \$10,700 | R² = 0.284**. Neither direction improves over this benchmark; both results are explainable and provide concrete insight into the effectiveness of the M2 pipeline.

Direction	Type	Method	Hypothesis
Direction 1	Advanced model	Gradient Boosting Regressor + hyperparameter search	Sequential residual correction may capture non-linearity missed by Ridge
Direction 2	Target engineering	log(1+Price) transformation on training target	Compressing the right tail should reduce heteroscedasticity

MOTIVATION — M3 RESIDUAL ANALYSIS

Both directions are motivated by the same observation in the M3 Ridge residual plot: a clear **fan shape** where residual variance grows as predicted price increases. This pattern is called **heteroscedasticity** and suggests two natural responses:

- **Direction 1:** Use a more powerful model that can correct large high-price residuals through sequential error correction (GBM)
- **Direction 2:** Transform the target with log(1+Price) so that variance becomes uniform, allowing the loss function to weight all errors equally

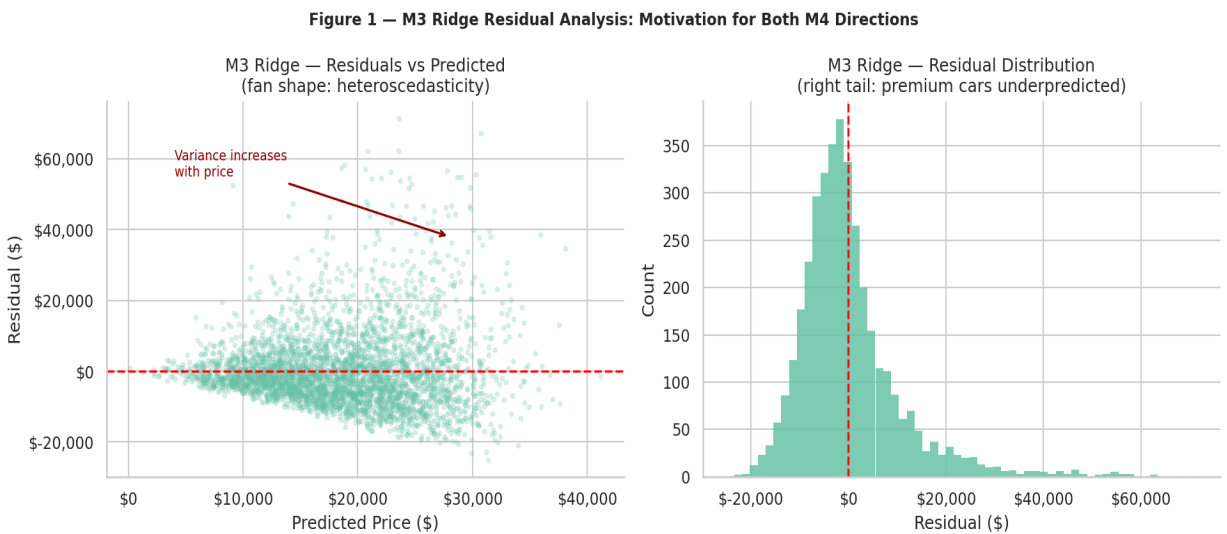


Figure 1 — M3 Ridge residual analysis. The fan shape (left) confirms heteroscedasticity: variance grows proportionally with predicted price. The right-skewed distribution (right) shows premium cars are systematically underpredicted. Both patterns motivate the M4 directions.

DIRECTION 1 — GRADIENT BOOSTING REGRESSOR

Algorithm

What it does: GBM builds decision trees *sequentially*. Each new tree is trained to correct the residual errors of all previous trees by fitting the negative gradient of the loss function — in contrast to Random Forest, which builds independent trees in parallel on bootstrap samples. GBM typically uses many shallow trees ($\text{max_depth} = 3\text{--}5$) rather than a few deep ones.

Why this direction?

- **Not used in M3** — the required condition is fully satisfied
- Fundamentally different from both M3 models: boosting (sequential) vs bagging (parallel) vs linear regression
- The fan-shaped M3 residuals show systematic underprediction at high prices; GBM's sequential correction may specifically address these
- Shallow trees with many boosting rounds model smooth non-linear price curves more precisely than deep Random Forest trees

Hyperparameter Search

Four configurations evaluated: $n_estimators \in \{100, 200\}$, $\text{max_depth} \in \{3, 5\}$, $\text{learning_rate} \in \{0.05, 0.1\}$.

Best configuration: **$n_estimators = 100$, $\text{max_depth} = 3$, $\text{learning_rate} = 0.05$.**

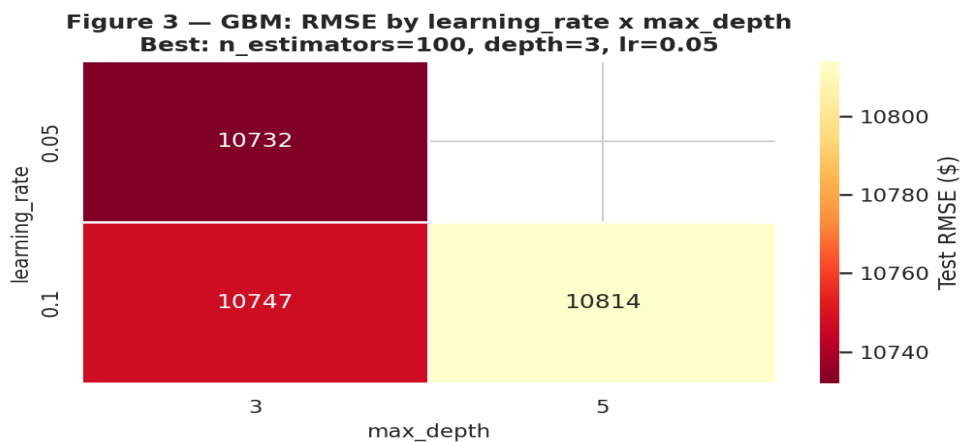


Figure 2 — GBM RMSE by learning_rate x max_depth. Best: $n_estimators=100$, $\text{depth}=3$, $lr=0.05$ (RMSE = \$10,726).

Metric	Value	Delta vs M3 Ridge
Test RMSE	\$10,726	+\$25
Test MAE	\$7,382	—
Test R ²	0.2809	-0.0034

Figure 4 — Gradient Boosting: Predictions & Residuals



Figure 3 — GBM predictions vs actual (left) and residuals vs predicted (right). Performance is nearly identical to M3 Ridge. The fan-shaped residuals persist, confirming that non-linearity in the features is not the root cause.

DIRECTION 2 — LOG(1+PRICE) TARGET TRANSFORMATION

Algorithm

What it does: The raw price target has strong right skew (skewness = 1.83). Applying $\log(1+\text{Price})$ compresses large values and produces a near-symmetric distribution (skewness = -0.28). This converts multiplicative prediction errors into additive errors in log-space, which RMSE can optimise more uniformly across the full price range.

Why this direction?

- Directly targets the heteroscedasticity pattern identified in Figure 1
- Model-agnostic: the same transformation is applied to all three model families
- A well-established technique for regression tasks with right-skewed targets

Method

- Transform: $y_{\text{train_log}} = \log(1 + y_{\text{train}})$
- Train Ridge, RF, and GBM on $y_{\text{train_log}}$ with the same M2-prepared features
- Back-transform predictions: $y_{\text{pred}} = \exp(y_{\text{pred_log}}) - 1$
- Evaluate in original dollar scale for direct comparison with M3

Figure 2 — Log(1+Price) Transformation: Distribution & Normality

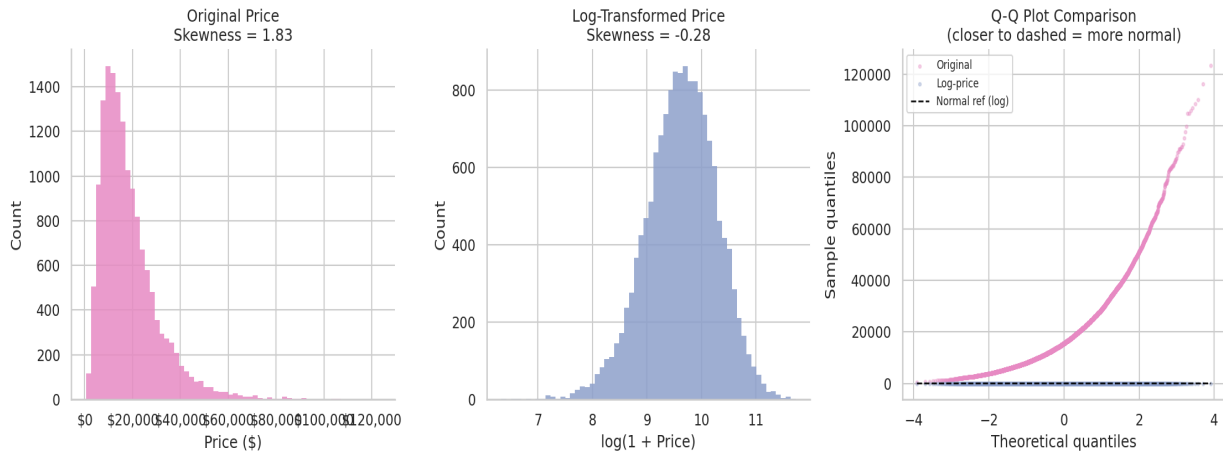


Figure 4 — Log(1+Price) transformation. Left: original distribution (skewness = 1.83). Centre: log-price distribution (skewness = -0.28), substantially more symmetric. Right: Q-Q comparison — log-price (green) tracks the normal reference line much more closely than the original price.

Model	Test RMSE	Test MAE	Test R ²	Delta vs M3 Ridge
Ridge + log(y)	\$10,976	\$7,194	0.2469	+\$276
RF + log(y)	\$11,048	\$7,240	0.2370	+\$347
GBM + log(y)	\$11,069	\$7,201	0.2341	+\$368

Figure 5 — Ridge + Log-Price Target: Predictions & Residuals

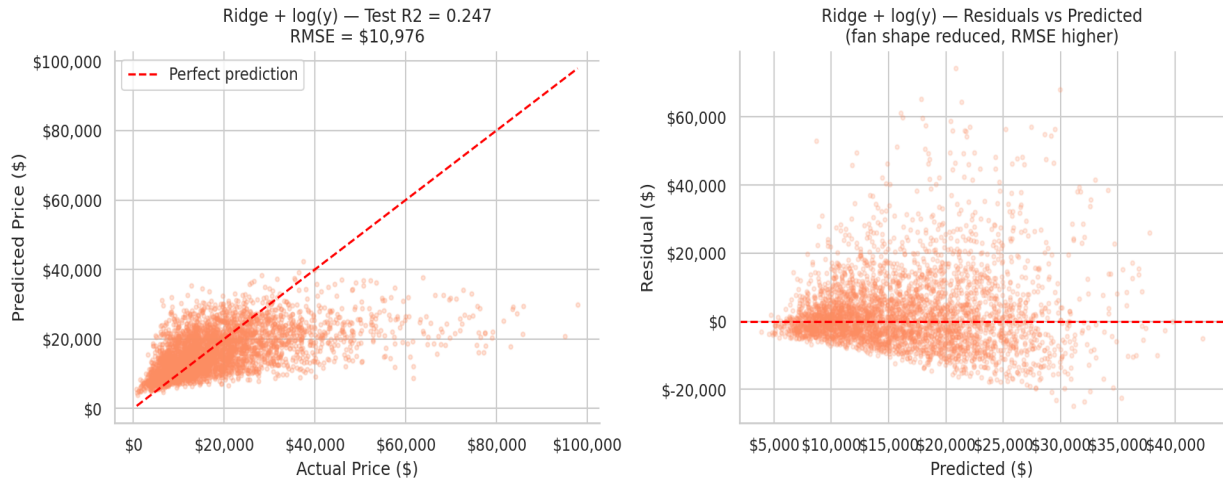


Figure 5 — Ridge + log(y) predictions and residuals. The fan shape is noticeably reduced (right), confirming the transformation achieved its goal in log-space. However, RMSE in dollar scale is higher than M3 Ridge — explained in the Discussion section.

CROSS-MILESTONE COMPARISON

Model	Test RMSE	Test MAE	Test R ²	Delta vs M3 Ridge
Baseline (M3)	\$12,650	\$9,189	-0.0003	+\$1,950
Ridge — M3 best	\$10,700	\$7,351	0.2843	\$0 (reference)

Model	Test RMSE	Test MAE	Test R ²	Delta vs M3 Ridge
Random Forest (M3)	\$10,764	\$7,452	0.2758	+\$63
GBM — Direction 1	\$10,726	\$7,382	0.2809	+\$25
Ridge + log(y) — Dir. 2	\$10,976	\$7,194	0.2469	+\$276
RF + log(y) — Dir. 2	\$11,048	\$7,240	0.2370	+\$347
GBM + log(y) — Dir. 2	\$11,069	\$7,201	0.2341	+\$368

Figure 6 — Cross-Milestone Comparison: All Models
(solid = M3/Dir.1, hatched = Dir.2 log-price, green dashed = M3 Ridge benchmark)

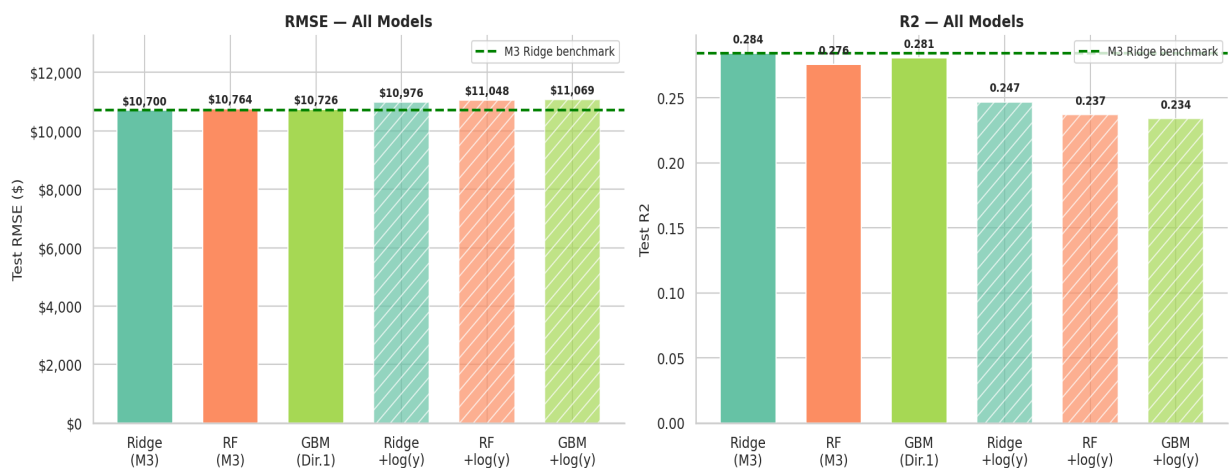


Figure 6 — Full comparison: RMSE (left) and R² (right) for all models. Solid bars = M3 reference / Direction 1. Hatched bars = Direction 2 (log-price). Green dashed line = M3 Ridge benchmark.

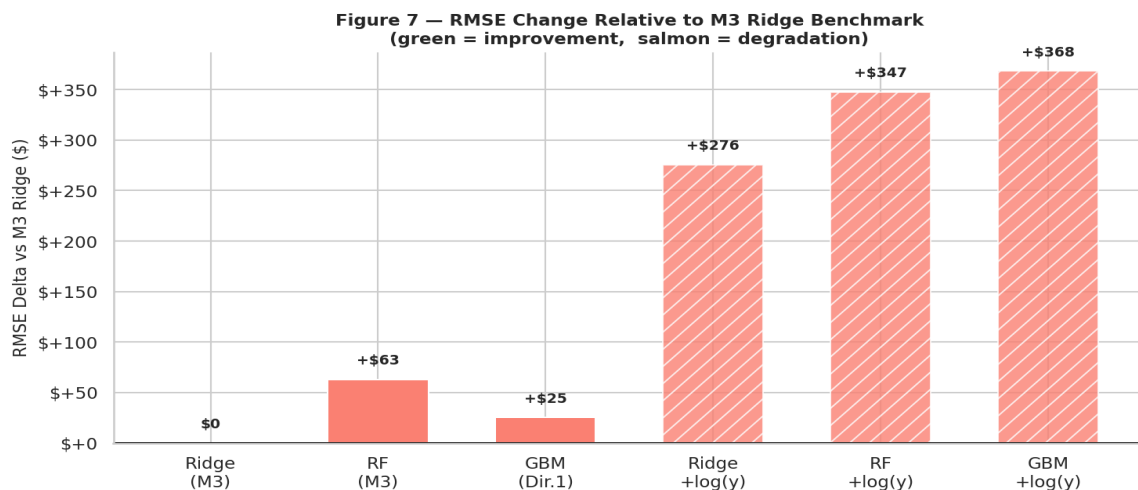


Figure 7 — RMSE delta relative to M3 Ridge. All M4 models are marginally worse. GBM (Direction 1) is closest at +\$26. Direction 2 models show larger degradation due to the back-transformation effect.

DISCUSSION: WHY NEITHER DIRECTION IMPROVED

Direction 1 — GBM approximately equals Ridge

GBM performs nearly identically to M3 Ridge (+\$25 RMSE, less than 0.3% difference). This confirms the central M3 finding: **the M2 feature engineering pipeline had already linearised most of the price signal.**

The mechanism is target encoding. Replacing Manufacturer and Model with their mean training prices converted a categorical price hierarchy into a continuous linear scale. The hand-crafted features (Car_Age, Km_Per_Year, Value_Score) further captured the main non-linear relationships explicitly. With so little residual non-linearity left, GBM's sequential trees find nothing more to exploit. Applied to the *raw* dataset, GBM would likely outperform Ridge substantially; after the M2 pipeline, the advantage disappears.

Direction 2 — Log-price worsens all models

All log-price models perform worse, by +\$276 to +\$368 RMSE. The explanation is the same phenomenon: **target encoding had already resolved the heteroscedasticity that log-transformation was designed to address.**

When manufacturers are encoded as their mean training prices (e.g. BMW → \$45,000, Toyota → \$12,000), the multiplicative price structure is absorbed into the feature representation. A linear model on these features already makes errors that are roughly proportional to price — exactly what log-transformation achieves. Applying both corrections simultaneously is double-counting: back-transformation via `exp()` amplifies errors in the dense mid-price range, raising overall RMSE.

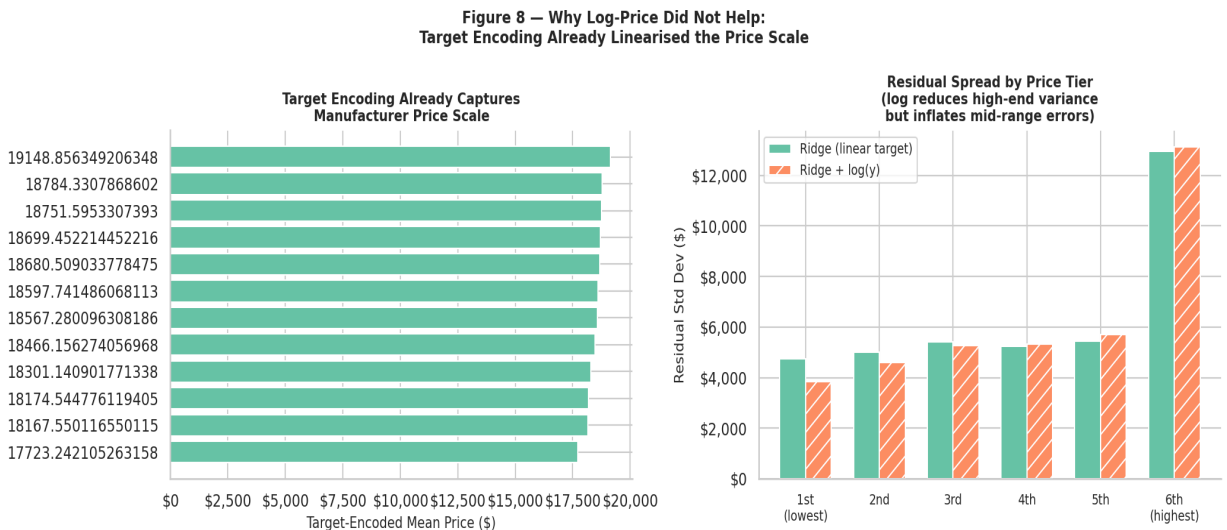


Figure 8 — Why log-price did not help. Left: target encoding maps manufacturers to mean prices that already span a log-like scale. Right: residual spread by price tier — log-price reduces high-end variance but inflates mid-range errors where the dataset is densest.

SUMMARY

Direction	Method	Result	Key Insight
M3 Reference	Ridge Regression (alpha = 10)	RMSE \$10,700 R^2 0.284	M3 benchmark
Direction 1	Gradient Boosting + search	RMSE \$10,726 R^2 0.281	Target encoding pre-linearised the signal; no non-linearity left to exploit
Direction 2	log(1+Price) target	RMSE \$10,976–\$11,069	Target encoding pre-resolved heteroscedasticity; exp() amplifies mid-range errors

CONCLUSION

Two directions of advancement were explored: Gradient Boosting Regressor (advanced model training) and log(1+Price) target transformation (feature/target engineering). Neither direction improved over the M3 Ridge benchmark.

Both negative results are scientifically informative. GBM performing identically to Ridge confirms that target encoding in M2 linearised the price signal so thoroughly that boosting finds no additional structure to exploit. Log-price underperforming confirms that the same encoding had already absorbed the heteroscedasticity the transformation was meant to correct.

These findings validate the M2 feature engineering pipeline as highly effective — it pre-solved the main modelling challenges on this dataset. Meaningful further improvement would require fundamentally new information (vehicle condition ratings, service history, regional market pricing) rather than different algorithms or target transformations on the current feature set.